Linear Algebra – Problem Set 5

Column, Row & Null Spaces · Basis, Dimension, Rank & Nullity · 13 problems · Original problems in the spirit of MIT 18.06 (Strang)

Set 4 asked what the words *mean*. This set asks what they **compute to**. Every matrix A carries three subspaces — $\mathrm{Col}(A)$, $\mathrm{Row}(A)$ and $\mathrm{Nul}(A)$ — and the whole machinery of row reduction exists to describe them. One fact organizes everything: **row operations preserve the row space and the dependence pattern of the columns, but not the column space itself.** That single asymmetry explains why the basis for $\mathrm{Col}(A)$ is taken from the original matrix while the basis for $\mathrm{Row}(A)$ is taken from the reduced one (Problems 9, 10). The set ends with rank–nullity, which converts "shape plus one number" into answers about pivots, independence and solvability (Problems 11–13).

Problem 1

Two sets in $\mathbb{R}^3$ differ by a single number:

Problem 2

Let $S = \mathrm{Span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$ and $T = \mathrm{Span}\left\{\begin{bmatrix}1\\-1\end{bmatrix}\right\}$, two lines through the origin in $\mathbb{R}^2$.

Problem 3

Let A be a 5×3 matrix. Without knowing any entry of A, state the ambient vector space containing each of $\mathrm{Col}(A)$, $\mathrm{Row}(A)$ and $\mathrm{Nul}(A)$, and justify each answer by counting entries.

Problem 4

Let $A = \begin{bmatrix}1 & 2 & 0 & 1\\0 & 1 & 1 & 2\\1 & 3 & 1 & 3\end{bmatrix}$, and write its columns as a_1, a_2, a_3, a_4.

Problem 5

Let $A = \begin{bmatrix}1 & 2 & 0\\0 & 1 & 1\\1 & 3 & 1\\2 & 5 & 1\end{bmatrix}$.

Problem 6

For each subset of $M_{2\times 2}$ below, decide whether it is a **subspace**. For those that are, prove it with the subspace test. For those that are not, name the first condition that fails and give an explicit counterexample (actual matrices, with numbers).

Problem 7

Consider the four vectors $w_1 = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $w_2 = \begin{bmatrix}0\\1\\2\end{bmatrix}$, $w_3 = \begin{bmatrix}1\\1\\3\end{bmatrix}$, $w_4 = \begin{bmatrix}2\\-1\\0\end{bmatrix}$, and let $H = \mathrm{Span}\{w_1, w_2, w_3, w_4\}$.

Problem 8

Let $W = \{p(x) \in P_3 : p(1) = 0\}$, the polynomials of degree at most 3 that vanish at $x = 1$.

Problem 9

Let $A = \begin{bmatrix} 2 & 1 & 0 & 1 & 3 \\ 4 & 2 & 1 & 2 & 5 \\ 1 & 0 & 1 & 1 & 1 \\ 3 & 1 & 1 & 2 & 4 \end{bmatrix}$. Row reduction gives

Problem 10

A student writes: *"After row reducing A to R, I should use the pivot columns of R as a basis for* $\mathrm{Col}(A)$*, since A and R are row equivalent."*

Problem 11

Suppose A is a 6×9 matrix with $\mathrm{rank}(A) = 4$.

Problem 12

Let $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ in $\mathbb{R}^4$.

Problem 13

Let A be an $n \times n$ matrix.